

What is Trigger?

- A Trigger is a database object that **automatically executes (fires) when a specific event occurs on a table.**
- Interview Points
 - Triggers execute automatically.
 - They are associated with a table.
 - Trigger events are INSERT, UPDATE, DELETE.
 - Can execute BEFORE or AFTER an event.
 - Useful for auditing and data validation.
- Trigger are set of structure Query language (SQL) statement that perform particular task. They invoke specific event (after Insert,u,d – before I,u,d)
- Database triggers are sets of commands that get executed when an event (Before Insert, After Insert, On Update, on delete of a row) occurs on a table.
- Triggers are special type of stored procedures that are defined to execute automatically in place or after data modification.
- Trigger allows you to execute a batch of SQL code an insert, update or delete command is execute against a specific table.

```
CREATE OR REPLACE TRIGGERRESTRICT_EMP
BEFORE INSERT ON EMP
BEGIN
RAISE_APPLICATION_ERROR(-20987,'INSERT MAT KAR NAHI HOGA');
END;
```

Why Use Triggers?

- Maintain audit logs
- Track changes in data
- Enforce business rules
- Automatically update related tables

Types of Triggers

1. BEFORE INSERT
2. AFTER INSERT
3. BEFORE UPDATE
4. AFTER UPDATE
5. BEFORE DELETE
6. AFTER DELETE

Example of trigger AFTER INSERT

- Suppose we want to log every new employee added.

```
-- create employee table
CREATE TABLE employees (
    id INT,
    name VARCHAR(50)
);

-- employee_log table
CREATE TABLE employee_log (
    log_message VARCHAR(100)
);
```

```
-- create Trigger after emp insert
CREATE TRIGGER trg_after_employee_insert
AFTER INSERT ON employees
FOR EACH ROW
INSERT INTO employee_log
VALUES (CONCAT('Employee Added: ', NEW.name));

-- insert data
INSERT INTO employees VALUES (1, 'Mohit');

-- MySQL automatically inserts:    into employee_log.
Employee Added: Mohit
```

UPDATE Trigger Example (Employee Salary Update)

```
-- salary log table
CREATE TABLE salary_log (
    emp_id INT,
    old_salary DECIMAL(10,2),
    new_salary DECIMAL(10,2)
);

-- create trigger
CREATE TRIGGER trg_salary_update
AFTER UPDATE ON employees
FOR EACH ROW
INSERT INTO salary_log (
    emp_id,
    old_salary,
    new_salary
)
VALUES (
    OLD.id,
    OLD.salary,
    NEW.salary
);

-- now updte salary
UPDATE employees
SET salary = 60000
WHERE id = 1;

-- Trigger ke andar values
-- Update se pehle:
id = 1
salary = 50000

-- update ke baad
id = 1
salary = 60000
```

```
+-----+-----+-----+
| emp_id | old_salary | new_salary |
+-----+-----+-----+
| 1      | 50000     | 60000     |
+-----+-----+-----+
```